

Logistics System API Documentation

Architecture: Microservices

Version: 1.0



1. Security & Authentication Architecture

To ensure high security and performance across the microservices, the system implements **Asymmetric Encryption (RSA)** for Token Validation and inter-service communication:

1. **Private/Public Key Pair:**
 - The **Auth & Identity Service** holds the **Private Key**. It is the *only* service authorized to sign and generate JWTs (JSON Web Tokens).
 - All other services (User, Core, Inventory, Support) hold the **Public Key**.
2. **Token Verification:**
 - When a request reaches the API Gateway, or any specific microservice, the service uses the **Public Key** to decrypt and verify the JWT signature locally. This eliminates the need to send an HTTP request to the Auth service for every validation, preventing network bottlenecks.
3. **Internal Communication:** * Services communicate securely using Internal API Keys or by propagating the user's validated JWT in the Authorization header.



System Roles (Access Control)

- **ADMIN:** Full system control. Manages accounts, branches, profiles, and views financial reports.
- **EMPLOYEE:** Manages the operational flow. Updates shipment/pickup statuses, manages inventory, and handles complaints.
- **MERCHANT (Customer):** Creates pickup requests, manages their products, views their own shipments, wallet balance, and submits complaints.
- **COURIER: NO SYSTEM ACCESS.** Couriers do not have **Accounts**. They exist solely as **Profiles** in the database. **EMPLOYEE** and **ADMIN** roles assign tasks to couriers and manually update the statuses based on physical interactions or phone calls with the couriers.



2. API Endpoints Summary Tables

2.1 Auth & Identity Service

Method	Endpoint	Allowed Roles	Description
POST	/api/auth/login	All	Authenticate user and issue JWT
POST	/api/auth/validate	Internal Gateway	Validate JWT using Public Key
POST	/api/auth/accounts	ADMIN	Create a new user account
PUT	/api/auth/accounts/{id}	ADMIN, Self	Update account details (Password/Email)
DELETE	/api/auth/accounts/{id}	ADMIN	Soft delete an account

2.2 User & Branch Service

Method	Endpoint	Allowed Roles	Description
POST	/api/users/profiles	ADMIN	Create a profile (Employee, Courier, Merchant)
PUT	/api/users/profiles/{id}	ADMIN, Self	Update profile data (Name, Phone)
GET	/api/users/profiles	ADMIN, EMPLOYEE	View/Filter profiles
POST	/api/branches	ADMIN	Add a new branch
PUT	/api/branches/{id}	ADMIN	Update branch data
DELETE	/api/branches/{id}	ADMIN	Soft delete a branch
GET	/api/branches	ADMIN, EMPLOYEE	List all branches
GET	/api/branches/search	ADMIN, EMPLOYEE	View/Filter branches

2.3 Inventory Service

Method	Endpoint	Allowed Roles	Description
POST	/api/inventory/products	ADMIN, EMP, MERCHANT	Define a base product
PUT	/api/inventory/products/{id}	ADMIN, EMPLOYEE	Update product details
POST	/api/inventory/items/receive	ADMIN, EMPLOYEE	Receive physical items into stock (Item-Level)
GET	/api/inventory/items	All	Track inventory levels (Filter)

2.4 Core Logistics & Finance Service

Method	Endpoint	Allowed Roles	Description
POST	/api/pickups	MERCHANT	Merchant creates a pickup request
PUT	/api/pickups/{id}	ADMIN, EMPLOYEE	Update pickup details
PUT	/api/pickups/{id}/assign	ADMIN, EMPLOYEE	Assign a Courier to a pickup request
GET	/api/pickups/search	ADMIN, EMP, MERCHANT	View/Filter pickups history to track pickups
POST	/api/shipments	ADMIN, EMPLOYEE	Convert pickup to shipment & assign inventory & Courier
PUT	/api/shipments/{id}/status	ADMIN, EMPLOYEE	Update shipment status (Triggers finance update)
GET	/api/shipments/search	All	Search and filter shipments
POST	/api/finance/transactions	ADMIN, EMPLOYEE	Create manual financial transactions

GET	/api/finance/wallets	ADMIN, EMPLOYEE	See users' wallet balances
GET	/api/finance/wallets/ me	MERCHANT	Track own wallet balance
GET	/api/finance/transac tions	ADMIN, EMPLOYEE	View financial reports/transactions

2.5 Support & Notifications Service

Method	Endpoint	Allowed Roles	Description
POST	/api/complaints	MERCHANT	Submit a complaint regarding a shipment
PUT	/api/complaints/{id}/status	ADMIN, EMPLOYEE	Update complaint status and add reply notes
GET	/api/complaints/search	ADMIN, EMPLOYEE	View/Filter complaints history
GET	/api/complaints/me	MERCHANT	See opened complaints for the logged-in merchant
GET	/api/notifications/me	All	See personal notifications
POST	/api/notifications/trigger	Internal Services	Trigger WhatsApp/SMS/Email/Site notifications



3. Detailed API Specifications

Note: All API errors follow a standard JSON response format to ensure consistency across the microservices.

3.1 Auth & Identity Service

3.1.1 Login

- **Description:** Authenticates a user and generates an RSA-signed JWT.
- **Endpoint:** POST /api/auth/login
- **Roles:** All (Except Courier)
- **Request Body:**
 - {
 - "email": "admin@logistic.com",
 - "password": "secure123"
 - }
- **Response (200 OK):**
 - {
 - "token": "eyJhbGciOiJSUzI1NiIs...",
 - "role": "ADMIN",
 - "accountId": 1
 - }
- **Expected Errors:**
 - 400 Bad Request:
 - { "status": 400, "error": "Bad Request", "message": "Validation failed: invalid email format" }
 -
 - 401 Unauthorized:
 - { "status": 401, "error": "Unauthorized", "message": "Incorrect email or password" }
 -
 - 403 Forbidden:
 - { "status": 403, "error": "Forbidden", "message": "Account is inactive or soft-deleted" }
 -
 - 500 Internal Server Error:
 - { "status": 500, "error": "Internal Server Error", "message": "Cryptography/Token generation failure" }
 -

3.1.2 Validate Token

- **Description:** Internal endpoint used by API Gateway to verify JWT signature using the Public Key.
- **Endpoint:** POST /api/auth/validate
- **Roles:** INTERNAL ONLY
- **Request Body:** {"token": "eyJhbGciOiJ..."}
- **Response (200 OK):** {"isValid": true, "role": "ADMIN", "accountId": 1}
- **Expected Errors:**
 - 401 Unauthorized:
- {"status": 401, "error": "Unauthorized", "message": "Invalid, tampered, or expired token" }

3.1.3 Create Account

- **Description:** Admin creates a new login account for {" customer ", "employee", "Admin" }
- **Endpoint:** POST /api/auth/accounts
- **Roles:** ADMIN
- **Request Body:**
 - {
 - "email": "merchant@store.com",
 - "password": "tempPassword",
 - "role": "CUSTOMER"
 - }
- **Response (201 Created):** {"accountId": 5, "message": "Account created successfully"}
- **Expected Errors:**
 - 400 Bad Request:
- {"status": 400, "error": "Bad Request", "message": "Invalid email format or weak password" }
- -
 - 403 Forbidden:
- {"status": 403, "error": "Forbidden", "message": "User lacks ADMIN role" }
- -
 - 409 Conflict:
- {"status": 409, "error": "Conflict", "message": "Email already exists in the system" }

3.1.4 Update Account

- **Description:** Allows updating password or email. A user can update their own account, and Admin can update any account.
- **Endpoint:** PUT /api/auth/accounts/{id}
- **Roles:** ADMIN, Self (Token accountId must match path {id})
- **Request Body:**
 - {
 - "email": "newemail@store.com",
 - "password": "newSecurePassword"
 - }
- **Response (200 OK):** {"message": "Account updated successfully"}
- **Expected Errors:**
 - 400 Bad Request:
 - { "status": 400, "error": "Bad Request", "message": "Invalid data provided" }
 - 403 Forbidden:
 - { "status": 403, "error": "Forbidden", "message": "Attempting to edit another user's account without ADMIN privileges" }
 - 404 Not Found:
 - { "status": 404, "error": "Not Found", "message": "Account ID does not exist" }
 - 409 Conflict:
 - { "status": 409, "error": "Conflict", "message": "New email is already taken" }

3.1.5 Delete Account (Soft Delete)

- **Description:** Deactivates an account without removing it from the database entirely.
- **Endpoint:** DELETE /api/auth/accounts/{id}
- **Roles:** ADMIN
- **Response (200 OK):** {"message": "Account deactivated"}
- **Expected Errors:**
 - 403 Forbidden:
 - { "status": 403, "error": "Forbidden", "message": "User lacks ADMIN role" }
 - 404 Not Found:
 - { "status": 404, "error": "Not Found", "message": "Account ID does not exist" }

3.2 User & Branch Service

3.2.1 Create Profile

- **Description:** Creates personal details. Used for Employees, Merchants, and importantly, Couriers (who have `accountId = null`).
- **Endpoint:** `POST /api/users/profiles`
- **Roles:** ADMIN
- **Request Body:**
 - {
 - "fullName": "Ahmed Ali",
 - "phoneNumber": "01012345678",
 - "type": "COURIER",
 - "accountId": null,
 - "branchId": 2
 - }
- **Response (201 Created):** {`"profileId": 102, "message": "Profile created"`}
- **Expected Errors:**
 - 400 Bad Request:
 - {`"status": 400, "error": "Bad Request", "message": "Missing mandatory fields"`}
 -
 - 403 Forbidden:
 - {`"status": 403, "error": "Forbidden", "message": "Unauthorized access"`}
 -
 - 404 Not Found:
 - {`"status": 404, "error": "Not Found", "message": "Associated branchId or accountId not found"`}
 -
 - 409 Conflict:
 - {`"status": 409, "error": "conflict", "message": "profile already exists"`}

3.2.2 Update Profile

- **Description:** Update profile info. Users can update their own profile; Admins can update any.
- **Endpoint:** PUT /api/users/profiles/{id}
- **Roles:** ADMIN, Self
- **Request Body:**
 - {
 - "fullName": "Ahmed Ali updated",
 - "phoneNumber": "01011112222"
 - }
- **Response (200 OK):** {"message": "Profile updated successfully"}
- **Expected Errors:**
 - 400 Bad Request:
 - {"status": 400, "error": "Bad Request", "message": "Validation errors" }
 -
 - 403 Forbidden:
 - {"status": 403, "error": "Forbidden", "message": "Attempting to edit another profile without ADMIN rights" }
 -
 - 404 Not Found:
 - {"status": 404, "error": "Not Found", "message": "Profile not found" }

3.2.3 Update Branch

- **Description:** Update existing branch data (name, location, phone).
- **Endpoint:** PUT /api/branches/{id}
- **Roles:** ADMIN
- **Request Body:**
 - {
 - "name": "Alexandria Main Branch",
 - "location": "Smouha",
 - "phone": "0123456789"
 - }
- **Response (200 OK):** {"message": "Branch updated successfully"}
- **Expected Errors:**
 - 400 Bad Request:
 - {"status": 400, "error": "Bad Request", "message": "Invalid inputs" }
 -
 - 403 Forbidden:
 - {"status": 403, "error": "Forbidden", "message": "Missing Admin role" }
 - 404 Not Found:
 - {"status": 404, "error": "Not Found", "message": "Branch ID does not exist" }

3.2.4 Delete Branch (Soft Delete)

- **Description:** Soft deletes a branch, ensuring historical data related to the branch remains intact.
- **Endpoint:** DELETE /api/branches/{id}
- **Roles:** ADMIN
- **Response (200 OK):** {"message": "Branch deactivated successfully"}
- **Expected Errors:**
 - 403 Forbidden:
- { "status": 403, "error": "Forbidden", "message": "Unauthorized" }
 -
 - 404 Not Found:
- { "status": 404, "error": "Not Found", "message": "Branch not found" }
 -
 - 409 Conflict:
- { "status": 409, "error": "Conflict", "message": "Cannot delete branch with active employees or active shipments" }
 -

3.2.5 View/Filter Branches

- **Description:** Retrieves a list of branches with optional filters (name, location, status).
- **Endpoint:** GET /api/branches/search
- **Roles:** ADMIN, EMPLOYEE
- **Response (200 OK):** [{ "id": 1, "name": "Main", ... }]
- **Expected Errors:**
 - 400 Bad Request:
- { "status": 400, "error": "Bad Request", "message": "Invalid filter parameters" }
 -
 - 403 Forbidden:
- { "status": 403, "error": "Forbidden", "message": "Missing permissions" }
 -

3.3 Inventory Service

3.3.1 Define Product

- **Description:** Defines a base product catalog item.
- **Endpoint:** POST /api/inventory/products
- **Roles:** ADMIN, EMPLOYEE, MERCHANT
- **Request Body:**
 - {
 - "merchantId": 105,
 - "name": "Black Polo Shirt XL",
 - "baseSku": "POLO-BLK-XL"
 - }
- **Response (201 Created):** {"productId": 10, "message": "Product defined"}
- **Expected Errors:**
 - 400 Bad Request:
 - { "status": 400, "error": "Bad Request", "message": "Missing data" }
 - 403 Forbidden:
 - { "status": 403, "error": "Forbidden", "message": "Merchant trying to create product for another Merchant ID" }
 - 409 Conflict:
 - { "status": 409, "error": "Conflict", "message": "baseSku already exists" }

3.3.2 Update Product Details

- **Description:** Modifies existing product details (e.g., name, base SKU).
- **Endpoint:** PUT /api/inventory/products/{id}
- **Roles:** ADMIN, EMPLOYEE
- **Request Body:**
 - {
 - "name": "Black Polo Shirt XL - Updated",
 - "baseSku": "POLO-BLK-XL-V2"
 - }
- **Response (200 OK):** {"message": "Product updated"}
- **Expected Errors:**
 - 400 Bad Request:
 - { "status": 400, "error": "Bad Request", "message": "Invalid data" }
 - 403 Forbidden:
 - { "status": 403, "error": "Forbidden", "message": "Unauthorized" }
 - 404 Not Found:
 - { "status": 404, "error": "Not Found", "message": "Product ID not found" }

3.3.3 Receive Inventory Items

- **Description:** When the Courier brings items from a pickup, the Employee scans them into the system, assigning unique serials.
- **Endpoint:** POST /api/inventory/items/receive
- **Roles:** ADMIN, EMPLOYEE
- **Request Body:**
 - {
 - "productId": 10,
 - "branchId": 2,
 - "pickupRequestId": 55,
 - "uniqueSkus": ["POLO-BLK-XL-001", "POLO-BLK-XL-002"]
 - }
- **Response (201 Created):** {"message": "Items received into inventory successfully"}
- **Expected Errors:**
 - 400 Bad Request:
 - { "status": 400, "error": "Bad Request", "message": "Validation failed" }
 -
 - 404 Not Found:
 - { "status": 404, "error": "Not Found", "message": "Product ID or Branch ID not found" }
 -
 - 409 Conflict:
 - { "status": 409, "error": "Conflict", "message": "One or more Unique SKUs already exist in the inventory" }
 -

3.4 Core Logistics & Finance Service

3.4.1 Create Pickup Request

- **Description:** Merchant requests a courier to pick up items.
- **Endpoint:** POST /api/pickups
- **Roles:** MERCHANT
- **Request Body:**
 - {
 - "pickupAddress": "10 Maadi Street, Cairo",
 - "pickupTime": "2026-05-20T14:00:00",
 - "note": "Call upon arrival"
 - }
- **Response (201 Created):** {"pickupId": 55, "message": "Pickup request created"}
- **Expected Errors:**
 - 400 Bad Request:
 - { "status": 400, "error": "Bad Request", "message": "Invalid date format or missing address" }
 -
 - 403 Forbidden:
 - { "status": 403, "error": "Forbidden", "message": "Not a Merchant account" }
 -

3.4.2 Update Pickup Details

- **Description:** Modify details of a pending pickup request and updating the status of the pickup request {"Pending", "Accepted", "Canceled", "Courier Assigned", "Completed" }
- **Endpoint:** PUT /api/pickups/{id}
- **Roles:** ADMIN, EMPLOYEE
- **Request Body:**
 - {
 - "pickupAddress": "15 Maadi Street, Cairo",
 - "pickupTime": "2026-05-21T10:00:00",
 - "note": "Updated note"
 - }
- **Response (200 OK):** {"message": "Pickup updated successfully"}
- **Expected Errors:**
 - 400 Bad Request:
 - {"status": 400, "error": "Bad Request", "message": "Invalid date formatting" }
 - 403 Forbidden:
 - {"status": 403, "error": "Forbidden", "message": "Unauthorized access" }
 - 404 Not Found:
 - {"status": 404, "error": "Not Found", "message": "Pickup request not found" }
 - 409 Conflict:
 - {"status": 409, "error": "Conflict", "message": "Cannot update a pickup that is already completed or canceled" }

3.4.3 Assign Courier to Pickup

- **Description:** Employees assign a specific courier profile to the pickup request.
- **Endpoint:** PUT /api/pickups/{id}/assign
- **Roles:** ADMIN, EMPLOYEE
- **Request Body:**
 - {
 - "courierProfileId": 102
 - }
- **Response (200 OK):** {"message": "Courier assigned successfully"}
- **Expected Errors:**
 - 400 Bad Request:
 - {"status": 400, "error": "Bad Request", "message": "Profile ID provided does not belong to a Courier" }
 - 404 Not Found:
 - {"status": 404, "error": "Not Found", "message": "Pickup request or Courier profile ID not found" }

3.4.4 View/Filter Pickup History

- **Description:** Retrieve pickups history to track operations. Merchants can only see their own pickups.
- **Endpoint:** GET /api/pickups/search
- **Roles:** ADMIN, EMPLOYEE, MERCHANT
- **Response (200 OK):** [{ "id": 55, "status": "PENDING", ... }]
- **Expected Errors:**
 - 400 Bad Request:
 - { "status": 400, "error": "Bad Request", "message": "Invalid filter parameters" }
 - 401 Unauthorized:
 - { "status": 401, "error": "Unauthorized", "message": "Invalid token" }
 - 403 Forbidden:
 - { "status": 403, "error": "Forbidden", "message": "Invalid role permissions" }

3.4.5 Create Shipment

- **Description:** Employees convert a picked-up request into an official shipment, linking specific inventory items to it.
- **Endpoint:** POST /api/shipments
- **Roles:** ADMIN, EMPLOYEE
- **Request Body:**
 - {
 - "trackingNumber": "TRK-987654321",
 - "customerId": 105,
 - "pickupRequestId": 55,
 - "codAmount": 600.00,
 - "shippingFee": 50.00,
 - "receiverName": "Mahmoud Hassan",
 - "receiverPhone": "01122334455",
 - "receiverAddress": "Nasr City, Cairo",
 - "inventoryItemIds": [501, 502]
 - }
- **Response (201 Created):** { "shipmentId": 300, "message": "Shipment created successfully" }
- **Expected Errors:**
 - 400 Bad Request:
 - { "status": 400, "error": "Bad Request", "message": "Missing mandatory shipment data" }
 - 404 Not Found:
 - { "status": 404, "error": "Not Found", "message": "Customer, Pickup, or Inventory item not found" }
 - 409 Conflict:
 - { "status": 409, "error": "Conflict", "message": "Tracking number already exists, or inventory item is already shipped/reserved" }

3.4.6 Update Shipment Status

- **Description:** Employees update status:
- {"DELIVERED", "Pending Pickup", "In Transit to Warehouse", "Arrived at Warehouse", "Out for Delivery", "In Transit to Customer"}. If **DELIVERED**, the system automatically handles wallet transactions.
- **Endpoint:** PUT /api/shipments/{id}/status
- **Roles:** ADMIN, EMPLOYEE
- **Request Body:**
 - {
 - "status": "DELIVERED"
 - }
 -
- **Response (200 OK):** {"message": "Shipment status updated"}
- **Expected Errors:**
 - 400 Bad Request:
- {"status": 400, "error": "Bad Request", "message": "Invalid status transition (e.g., pending directly to delivered)" }
- 404 Not Found:
- {"status": 404, "error": "Not Found", "message": "Shipment not found" }

3.4.7 Create Transaction (Manual Adjustment)

- **Description:** Admin/Employee can create manual transactions (e.g., adjustments, cash handovers).
- **Endpoint:** POST /api/finance/transactions
- **Roles:** ADMIN, EMPLOYEE
- **Request Body:**
 - {
 - "walletId": 88,
 - "amount": 100.00,
 - "type": "CREDIT",
 - "referenceId": null,
 - "notes": "Manual compensation adjustment"
 - }
- **Response (201 Created):** {"transactionId": 12, "message": "Transaction recorded"}
- **Expected Errors:**
 - 400 Bad Request:
- {"status": 400, "error": "Bad Request", "message": "Invalid amount or transaction type" }
- 403 Forbidden:
- {"status": 403, "error": "Forbidden", "message": "Missing permissions" }
- 404 Not Found:
- {"status": 404, "error": "Not Found", "message": "Wallet ID not found" }

3.4.8 View All Users' Wallets

- **Description:** Admin and Employees can view all wallets to monitor balances.
- **Endpoint:** GET /api/finance/wallets
- **Roles:** ADMIN, EMPLOYEE
- **Response (200 OK):** [{ "walletId": 88, "balance": 550.00 }]
- **Expected Errors:**
 - 403 Forbidden:
- { "status": 403, "error": "Forbidden", "message": "Only Admin or Employee can view all wallets" }
-

3.4.9 Track Own Balances

- **Description:** Gets wallet balance and latest transactions for the logged-in merchant.
- **Endpoint:** GET /api/finance/wallets/me
- **Roles:** MERCHANT
- **Response (200 OK):**
 - {
 - "walletId": 88,
 - "balance": 550.00,
 - "transactions": [
 - {
 - "amount": 600.00,
 - "type": "CREDIT",
 - "referenceId": 300,
 - "createdAt": "2026-04-14T10:00:00"
 - }
 -]
 - }
 -
- **Expected Errors:**
 - 404 Not Found:
- { "status": 404, "error": "Not Found", "message": "Wallet not initialized for this merchant" }
-

3.5 Support & Notifications Service

3.5.1 Submit Complaint

- **Description:** Merchants can submit a complaint, optionally linked to a specific shipment.
- **Endpoint:** POST /api/complaints
- **Roles:** MERCHANT
- **Request Body:**
 - {
 - "shipmentId": 300,
 - "subject": "Delayed Delivery",
 - "content": "The shipment has not arrived yet."
 - }
- **Response (201 Created):** {"complaintId": 15, "message": "Complaint submitted successfully"}
- **Expected Errors:**
 - 400 Bad Request:
 - { "status": 400, "error": "Bad Request", "message": "Subject or content missing" }
 - 404 Not Found:
 - { "status": 404, "error": "Not Found", "message": "Shipment ID provided does not exist" }

3.5.2 Update Complaint Status & Add Note

- **Description:** Employees resolve or update complaint status {"Pending", "In Progress", "Resolved", "Compensated"}, and leave a reply note for the merchant.
- **Endpoint:** PUT /api/complaints/{id}/status
- **Roles:** ADMIN, EMPLOYEE
- **Request Body:**
 - {
 - "status": "RESOLVED",
 - "note": "We apologize, the shipment was delivered today."
 - }
- **Response (200 OK):** {"message": "Complaint status updated"}
- **Expected Errors:**
 - 400 Bad Request:
 - { "status": 400, "error": "Bad Request", "message": "Invalid status update" }
 - 404 Not Found:
 - { "status": 404, "error": "Not Found", "message": "Complaint not found" }
- 400 Bad Request:
 - { "status": 400, "error": "Bad Request", "message": "Invalid status transition (e.g., pending directly to resolved)" }

3.5.3 View/Filter Complaints History

- **Description:** Admin and Employees can view and filter the history of all complaints.
- **Endpoint:** GET /api/complaints/search
- **Roles:** ADMIN, EMPLOYEE
- **Response (200 OK):** [{ "id": 15, "subject": "Delayed Delivery", "status": "RESOLVED" }]
- **Expected Errors:**
 - 403 Forbidden:
- { "status": 403, "error": "Forbidden", "message": "Merchant attempting to access the global complaints list" }
 -

3.5.4 View Own Opened Complaints

- **Description:** Merchant can retrieve their own opened/active complaints.
- **Endpoint:** GET /api/complaints/me
- **Roles:** MERCHANT
- **Response (200 OK):** [{ "id": 16, "subject": "Damaged item", "status": "OPEN" }]
- **Expected Errors:**
 - 401 Unauthorized:
- { "status": 401, "error": "Unauthorized", "message": "Token missing or invalid" }
 -

3.5.5 View Notifications

- **Description:** Any authenticated user can view their personal notifications.
- **Endpoint:** GET /api/notifications/me
- **Roles:** All
- **Response (200 OK):** [{ "message": "Shipment TRK-987654321 Delivered", "sentAt": "2026-04-14T10:00:00" }]
- **Expected Errors:**
 - 401 Unauthorized:
- { "status": 401, "error": "Unauthorized", "message": "User is not authenticated" }